Velocity and Acceleration as Derivatives

1. A particle moves in a straight line so that its displacement $x\ m$ from a fixed point O on the line at any time t seconds ($t \geq 0$) is given by $x = t^2 - 5t + 6$. Find:
   a. Its initial displacement
   b. Its initial velocity
   c. When it first passes through O and with what velocity
   d. When it passes through O the second time and with what velocity
   e. When and where its velocity is zero.
2. The displacement $x\ m$ at time t seconds ($t \geq 0$) of a particle moving in a straight line is given by $x = 2t^3 - t^2 + 4t + 1$. Find acceleration.
3. The displacement $x\ m$ at time t seconds ($t \geq 0$) of a particle moving in a straight line is given by $x = t^2 - 5t + 4$.
   a. At what time is its velocity zero?
   b. What is the acceleration at this time?
   c. What is the distance travelled in the first 4 seconds?
   d. At what time is the velocity $8\ ms^{-1}$
4. A point moving in a straight line is distant $x\ m$ from the origin O at time t, where $x = 2t^3 - 15t^2 + 36t$.
   a. Find the velocity and acceleration at any time t.
   b. Find the initial velocity and acceleration.
   c. At what times is the velocity zero?
   d. At what time is the acceleration zero? Find the velocity and position at this time.
   e. During what interval of time is the velocity negative?
5. The displacement $x\ m$ at time t seconds of a particle moving in a straight line is given by $x = 2t^3 - 9t^2 + 12t + 6$. Find:
   a. When its acceleration is zero, and its velocity at this time.
   b. When its velocity is zero, and its acceleration at this time.
6. Two bodies move along a straight path, starting at the same time, so that their displacement $x\ m$ from a fixed point O at any time t is given by $x = t + 6$ and $x = t^2 + 4$ respectively. At what times are they:
   a. Together
   b. Travelling with the same velocity?
7. The velocity $v\ ms^{-1}$ at time t seconds ($t \geq 0$) of a body moving in a straight line is given by $v = 6t^2 + 6t - 12$. Its initial displacement is $7\ m$ from O. Find:
   a. The acceleration when the velocity is zero
   b. The initial velocity and acceleration.
8. Two cars A and B travel along a straight road in the same direction. Their respective distances $x\ km$ from a fixed point O at any time t hours are given by the following rules:
   $A: x = 50t - 20t^2, B: x = 80t^2 + 20t$.
   a. Calculate each car's speed at the point O.
   b. At what time are the cars travelling at the same speed?
   c. Both cars reach a point Q at the same time. Calculate the distance from O to Q.
   d. A third car, travelling at uniform speed, is $2\ km$ ahead of A and B when they pass the point O. If this car arrives at Q at the same time as A and B, find a rule connecting x and t for it.
9. A particle moves in a straight line so that its displacement $x(t)$ from a foxed point in the line at time $t \geq 0$ is given by $x(t) = 3 + 4t - 5\sqrt{t^2 + 4}$. Find the particle's displacement when it comes to rest.

10. A particle is moving so that, for $0 < t < 1$, its velocity is positive and its acceleration is negative. Which graph could represent the displacement function of this particle?

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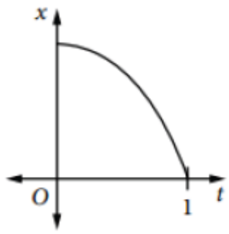

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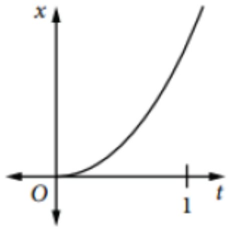

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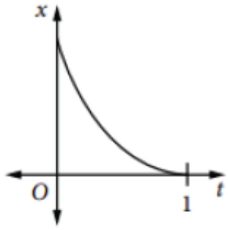

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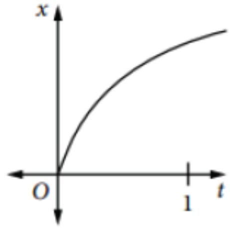
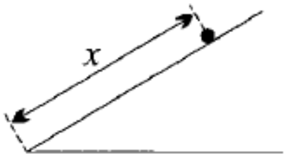


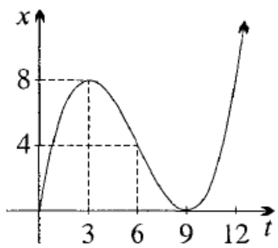
11. A particle moves according to the equation $x = t^2 - 8t$, in units of metres and seconds.
    a. Differentiate to find the functions v and $\ddot{x}$, and show that the acceleration is constant.
    b. What are the displacement, velocity and acceleration after 5 seconds?
    c. When is the particle stationary, and where is it then?
12. A particle moves on a horizontal line so that its displacement x cm to the right of the origin at time t seconds is $x = t^3 - 6t^2 - t + 2$.
    a. Differentiate to find v and $\ddot{x}$ as functions of t.
    b. Where is the particle initially, and what are its speed and acceleration?
    c. At time $t = 3$:
        i. Is the particle left or right of the origin?
        ii. Is it travelling to the left or to the right?
        iii. In what direction is it accelerating?
    d. When is the particle's acceleration zero, and what is its speed then?
13. A cricket ball is thrown vertically upwards, and its height x in metres at time t seconds after it is thrown is given by $x = 20t - 5t^2$.
    a. Find v and $\ddot{x}$ as functions of t, and show that the ball is always accelerating downwards. Then sketch graphs of x, v and $\ddot{x}$ against t.
    b. Find the speed at which the ball was thrown, find when it returns to the ground, and show that its speed then is equal to the initial speed.
    c. Find its maximum height above the ground, and the time to reach this height.
    d. Find the acceleration at the top of the flight, and explain why the acceleration can be nonzero when the ball is stationary.
    e. When is the ball's height 15 metres, and what are its velocities then?
14. A particle moves according to $x = t^2 - 8t + 7$, in units of metres and seconds.
    a. Find v and $\ddot{x}$ as functions of t, then sketch graphs of x, v and $\ddot{x}$ against t.
    b. When is the particle
        i. At the origin?
        ii. Stationary?
    c. What is the maximum distance from the origin, and when does it occur:
        i. During the first 2 seconds,
        ii. During the first 6 seconds,
        iii. During the first 10 seconds?
    d. What is the particle's average velocity during the first 7 seconds? When and where is its instantaneous velocity equal to this average?
    e. How far does it travel during the first 7 seconds, and what is its average speed?

15. A smooth piece of ice is projected up a smooth inclined surface, as shown to the right. Its distance x in metres up the surface at time t seconds is $x = 6t - t^2$.



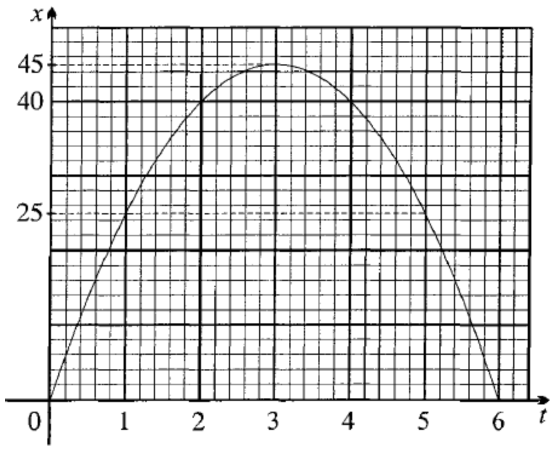
a. Find the functions v and $\ddot{x}$, and sketch x and v.
b. In which direction is the ice moving, and in which direction is it accelerating:
    i. When $t = 2$,
    ii. When $t = 4$?
c. When is the ice stationary, for how long is it stationary, where is it then, and is it accelerating then?
d. Find the average velocity over the first 2 seconds, and the time and place where the instantaneous velocity equals this average velocity.
e. Show that the average speed during the first 3 seconds, the next 3 seconds and the first 6 seconds are all the same.

16. A particle is moving horizontally so that its displacement x metres to the right of the origin at time t seconds is given by the graph to the right.



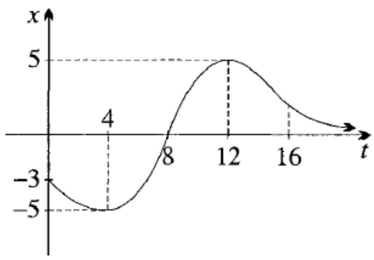
a. In the first 10 seconds, what is its maximum distance from the origin, and when does it occur?
b. When is the particle:
    i. Stationary,
    ii. Moving to the right,
    iii. Moving to the left?
c. When does it return to the origin, what is its velocity then, and in which direction is it accelerating?
d. When is its acceleration zero, where is it then, and in what direction is it moving?
e. During what time is its acceleration negative?
f. At about what time is:
    i. The displacement about the same as at $t = 2$?
    ii. The velocity about the same as at $t = 2$?
    iii. The speed about the same as at $t = 2$?
g. Sketch (roughly) the graphs of v and $\ddot{x}$.

17. A stone was thrown vertically upwards, and the graph to the right shows its height x metres at time t seconds after it was thrown.



a. What was the stone's maximum height, how long did it take to reach it, and what was its average speed during this time?
b. Draw tangents and measure their gradients to find the velocity of the stone at times $t = 0, 1, 2, 3, 4, 5$ and 6.
c. For what length of time was the stone stationary at the top of its flight?
d. The graph is concave down everywhere. How is this relevant to the motion?
e. Draw a graph of the instantaneous velocity of the stone from $t = 0$ to $t = 6$. What does the graph tell you about what happened to the velocity during these 6 seconds?

18. A particle is moving vertically according to the graph shown to the right, where upwards has been taken as positive.

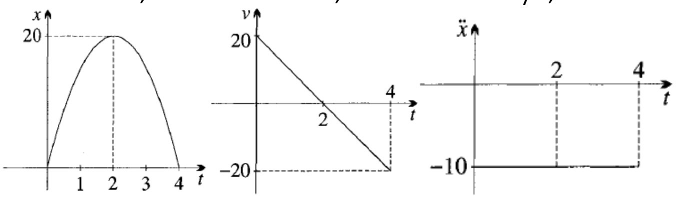


a. At what times is this particle
   i. Below the origin,
   ii. Moving downwards,
   iii. Accelerating downwards?
b. At about what time is its speed greatest?
c. At about what time is
   i. Distance from the origin about the same as at $t = 3$?
   ii. Velocity about the same as at $t = 3$?
   iii. Speed about the same as at $t = 3$?
d. How many times between $t = 4$ and $t = 12$ is the instantaneous velocity equal to the average velocity during this time?
e. How far will the particle eventually travel?
f. Sketch the graphs of v and $\ddot{x}$ as functions of time.

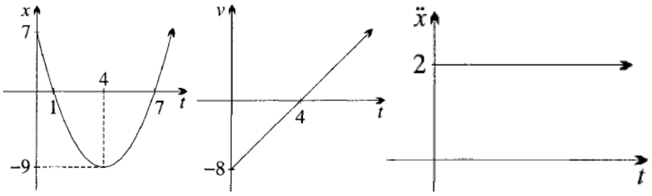
Answer

1. a. $x = 6\ m$; b. $v = -5\ ms^{-1}$; c. $t = 2s, v = -1\ ms^{-1}$; d. $t = 3s, v = 1\ ms^{-1}$; e. $t = 2.5\ s, x = -0.25\ m$
2. $a = 12t - 2$
3. a. $t = 2.5\ s$; b. $a = 2\ ms^{-2}$; c. $4 + 2 \times 2.25 = 8.5\ m$; d. $t = 6.5\ s$
4. a. $\dot{x} = 6t^2 - 30t + 36, \ddot{x} = 12t - 30$; b. $36\ ms^{-1}, -30\ ms^{-2}$; c. $t = 2, 3\ s$; d. $t = 2.5\ s, v = -1.5\ ms^{-1}, x = 27.5\ m$; e. $2 < t < 3$
5. a. $t = 1.5\ s, v = -1.5\ ms^{-1}$; b. $t = 1\ s, a = -6m\ s^{-2}$; $t = 2\ s, a = 6\ ms^{-2}$

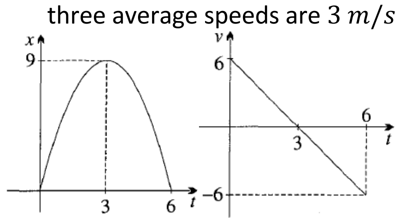
6. a. $t = 2\ s$; b. $t = 0.5\ s$
7. a. $t = 1, a = 18\ ms^{-2}$; b. $v = -12\ ms^{-1}, a = 6\ ms^{-2}$
8. a. $50\ km\ h^{-1}$, $20\ km\ h^{-1}$; b. $9\ min$; c. $13.2\ km$; d. $x = \frac{112t}{3} + 2$
9. $\dot{x} = 4 - \frac{5t}{\sqrt{t^2+4}}$, $t = 2\frac{2}{3}s$, $x = -3$
10. D
11. a. $v = 2t - 8$, $\ddot{x} = 2$, which is constant; b. $x = -15$ metres, $v = 2\ m/s$, $\ddot{x} = 2\ m/s^2$; c. when $t = 4$, $v = 0$ and $x = -16$.
12. a. $v = 3t^2 - 12t - 1$, $\ddot{x} = 6t - 12$; b. When $t = 0$, $x = 2\ cm$, $|v| = 1\ cm/s$, $\ddot{x} = -12\ cm/s^2$; c. i. left ($x = -28$), ii. left ($v = -10$), iii. right ($\ddot{x} = 6$); d. When $t = 2$, $\ddot{x} = 0$ and $|v| = 13\ cm/s$.
13. a. $x = 5t(4 - t)$, $v = 20 - 10t$, $\ddot{x} = -10$ (graph see below); b. It returns at $t = 4$; both speeds are $20\ m/s$; c. 20 metres after 2 seconds; d. $-10\ m/s^2$, although the ball is stationary, its velocity is changing, meaning that its acceleration is nonzero; e. after 1 second, when $v = 10\ m/s$, and after 3 seconds, when $v = -10\ m/s$.



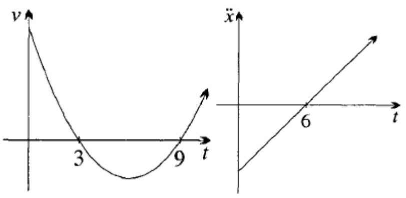
14. a. $x = (t - 7)(t - 1)$, $v = 2(t - 4)$, $\ddot{x} = 2$ (graph see below); b. i. $t = 1$ and $t = 7$, ii. $t = 4$; c. i. 7 metres when $t = 0$, ii. 9 metres when $t = 4$, iii. 27 metres when $t = 10$; d. $-1\ m/s$, $t = 3\frac{1}{2}$, $x = -8\frac{3}{4}$; e. 25 metres, $3\frac{4}{7}\ m/s$



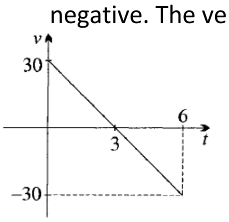
15. a. $x = t(6 - t)$, $v = 2(3 - t)$, $\ddot{x} = -2$ (graph see below); b. i. When $t = 2$, it is moving upwards and accelerating downwards, ii. When $t = 4$, it is moving downwards and accelerating downwards; c. $v = 0$ when $t = 3$, it is stationary for zero time, 9 metres up the plane, and is accelerating downwards at $2\ m/s^2$; d. $4\ m/s$, when $v = 4$, $t = 1$ and $x = 5$; e. all three average speeds are $3\ m/s$.



16. a. 8 metres when $t = 3$; b. i. when $t = 3$ and $t = 9$, ii. when $0 \leq t < 3$ and when $t > 9$, iii. When $3 < t < 9$; c. $t = 9$, $v = 0$, accelerating forwards; d. $t = 6$, $x = 4$, moving backwards; e. $0 \leq t < 6$; f. i. $t \doteqdot 4, 12$, ii. $t \doteqdot 10$, iii. $t \doteqdot 4, 8, 10$; g. see below



17. a. 45 metres, 3 seconds, $15\ m/s$; b. $30\ m/s$, 20, 10, 0, -10, -20, -30; c. 0 seconds; d. The acceleration was always negative. The velocity was decreasing at a constant rate of $10\ m/s$ every second; e. see below



18. a. i. $0 \leq t < 8$, ii. $0 \leq t < 4$ and $t > 12$, iii. Roughly $8 < t < 16$, b. roughly $t = 8$; c. i. $t \doteqdot 5, 11, 13$, ii. $t \doteqdot 13, 20$, iii. $t \doteqdot 5, 11, 13, 20$; d. twice; e. 17 units; f. see below